STRUCTURAL ANALYSIS 1 (INS2512) - LESSON 11 -

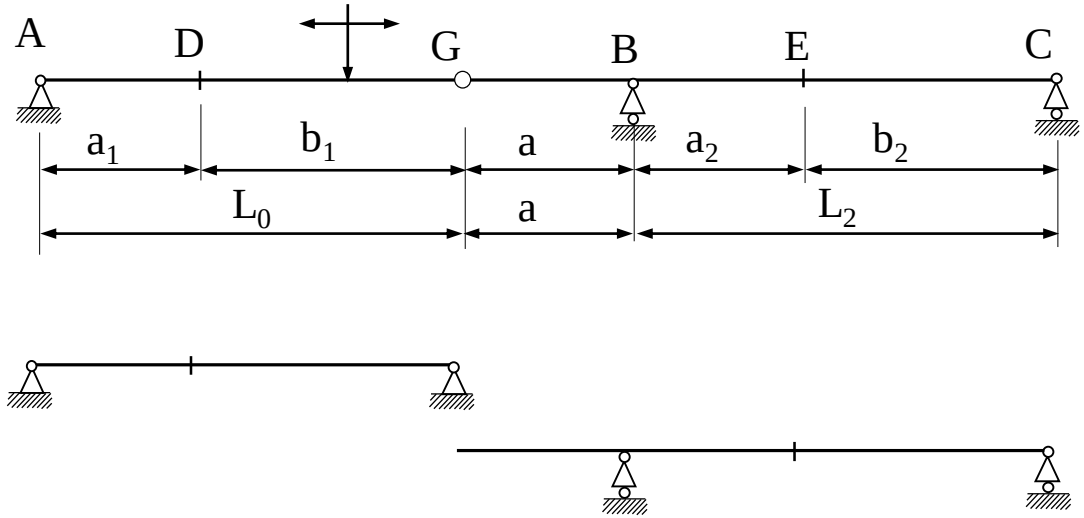
Assist. Prof. Dr. Muzaffer BÖREKÇİ
Email: structuralanalysis1mb@gmail.com

Influence Lines of Statically Determinate Systems

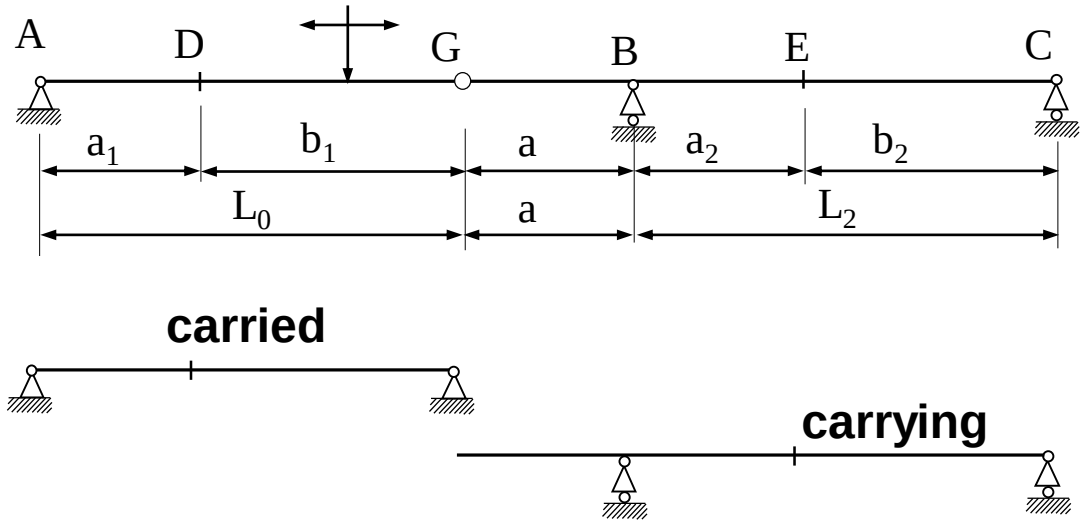
Gerber Beam

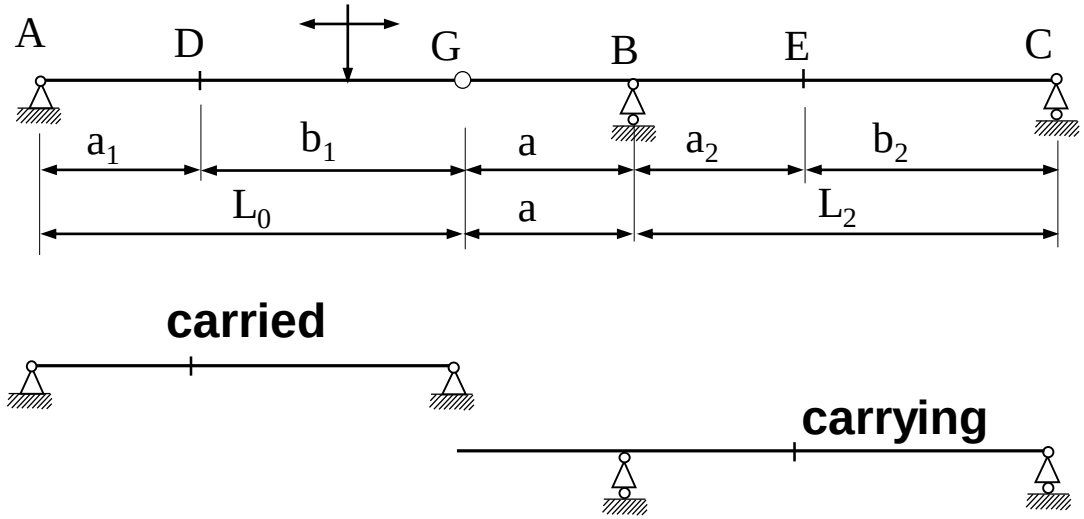
Influence line of a Gerber beam can be constructed as follows:

- The structural layout under vertical loads is drawn.



- While moving the unit vertical load on the structure, the regions in which the internal force or the support reaction, whose influence line will be constructed is zero, are determined by means of the structural layout.





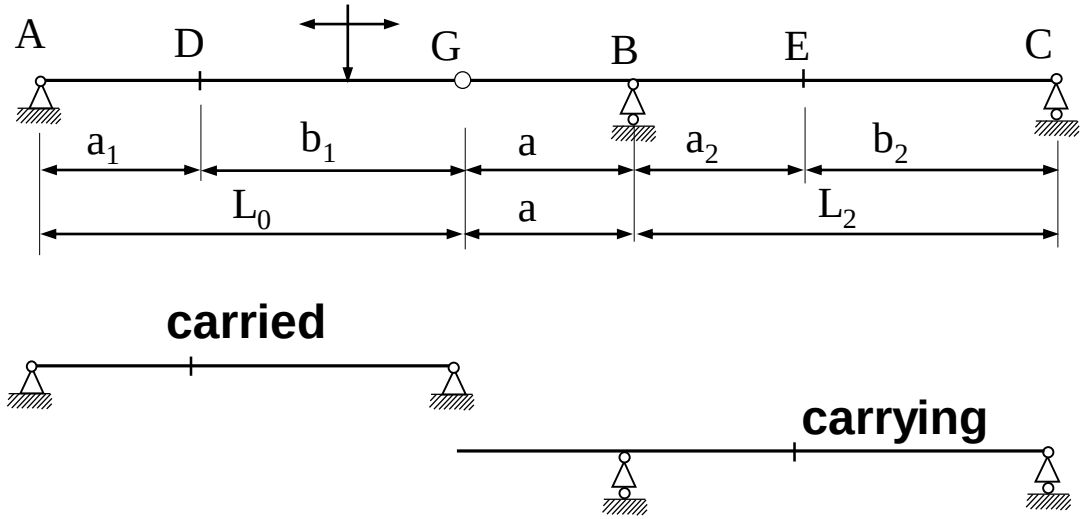
- **There is no load transfere from the carrying structure to the carried one.**

In other words, any internal force or support reaction in the carried structure is zero when the load acts on the carrying structure.

- The points at which the influence line of interest are zero are determined.

- **Zero** influence line points are supports in a Gerber beam.

- For the influence line of a vertical support reaction, the ordinate of the influence line at that support is **1**.



- The influence line of a support reaction / an internal force in the span of a Gerber beam where the influence line will be drawn is the same as influence line of simply supported beam, cantilever beam and overhanging beam.

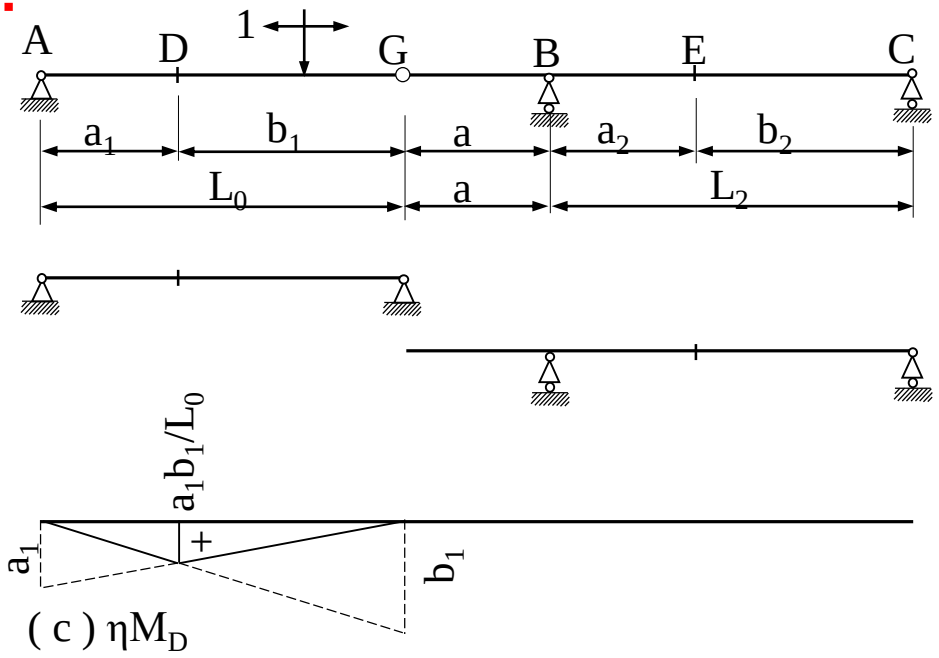
Hence, the influence lines of a Gerber beam can be obtained by:

- drawing the influence lines of simply supported beam, cantilever beam and overhanging beam

or

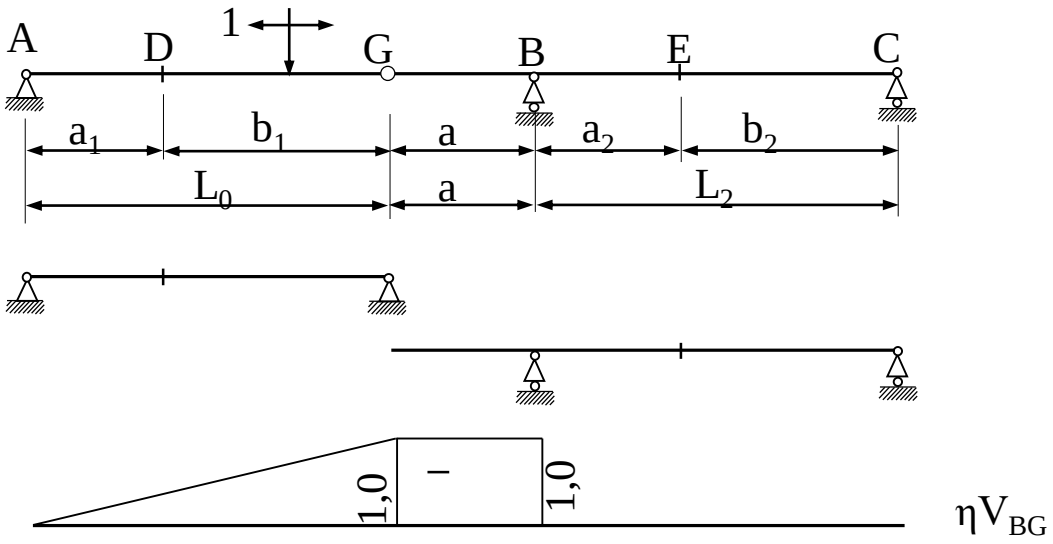
- extending the influence lines of simply supported beam until the end of overhanging parts.

- **Example 1:**



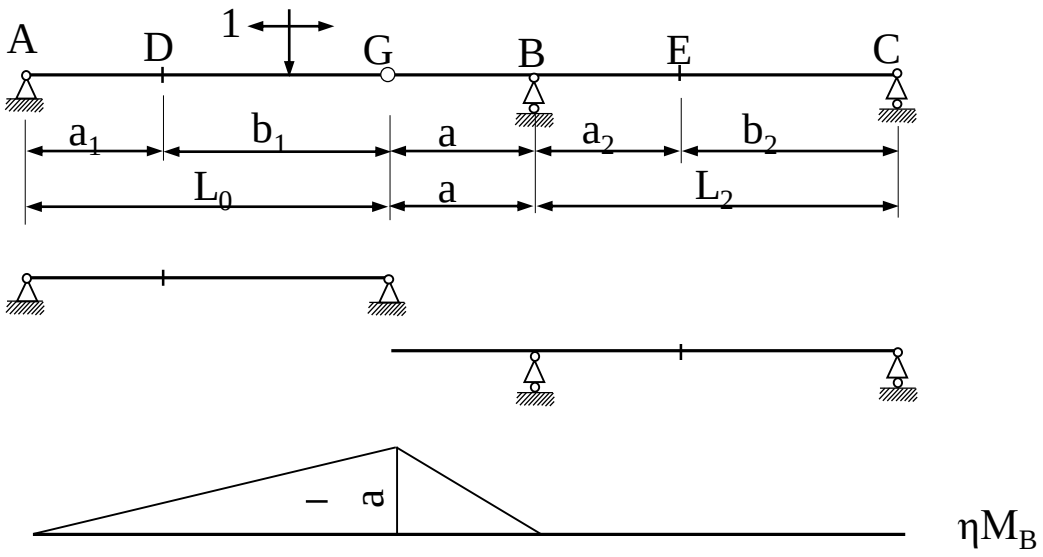
Influence line of bending moment at D (M_D):

- While the unit load moves in between G and C, it does not cause any bending moment at D.
- Then, the influence line of M_D is the same as that for a simply supported beam.



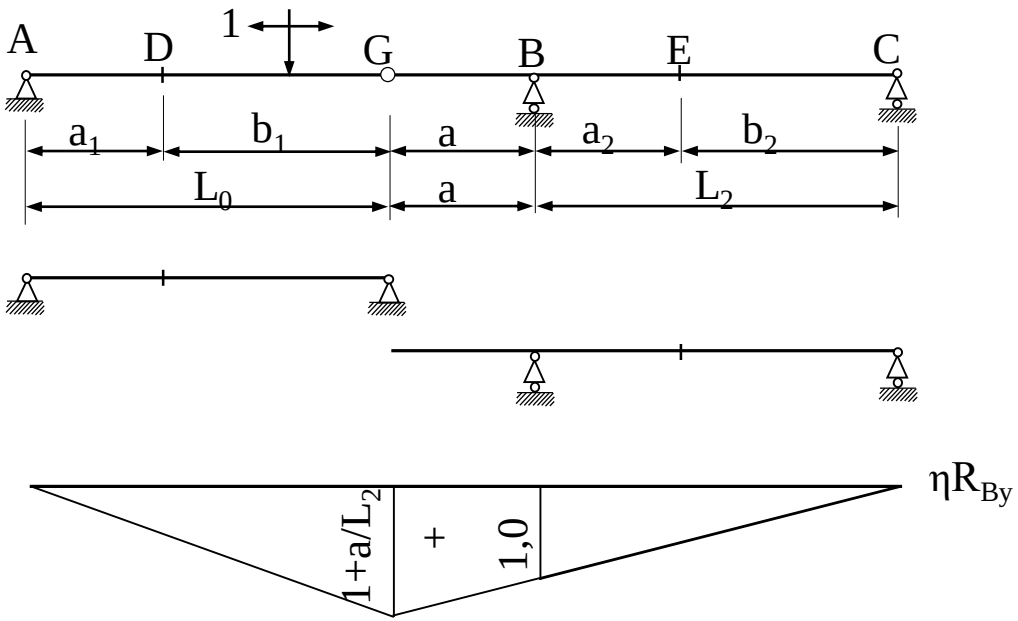
Influence line of shear force on the left of B (V_{BG}):

- While the unit load moves in between B and C, it does not cause any internal force on the left of B.
- When the unit load is at A, $V_{BG} = 0$.
- All ordinates in between G and B are -1 similar to a cantilever beam.



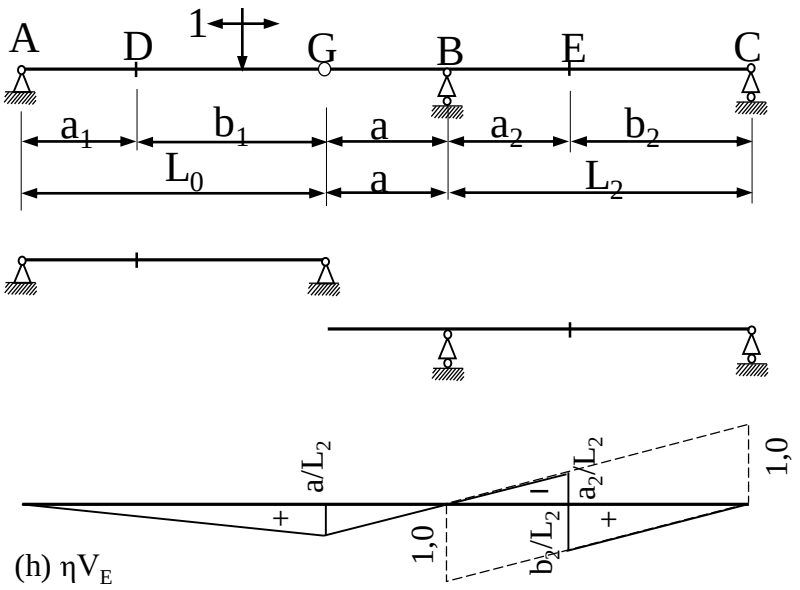
Influence line of bending moment at B (M_B):

- $M_{BG} = M_{BE} = M_B$ since there is no external moment.
- While the unit load moves in between B and C, when the unit load is at A or B, **$M_B = 0$.**
- **The influence line of M_B in between G and B** is equal to that for a cantilever beam.
- By connecting the ordinates at A and G, the influence line is constructed.



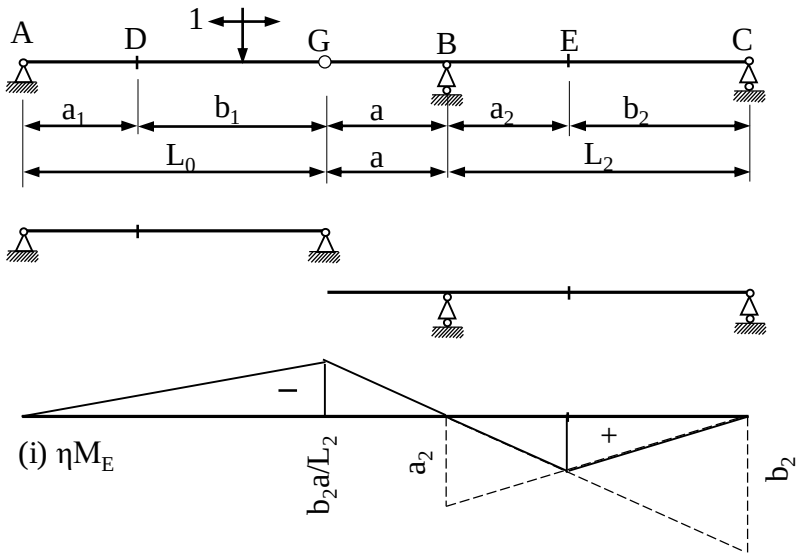
Influence line of vertical support reaction at B (R_{By}):

- While the unit load moves in between A and C, it causes R_{By} support reaction. In other words, there is zero region.
- When the load is at A and C, **$R_{By} = 0$.**
- First, the influence line for the simply supported beam of length L_2 is drawn, and the line is extended until G.
- Second, the ordinate at G is connected with the zero ordinate at A.



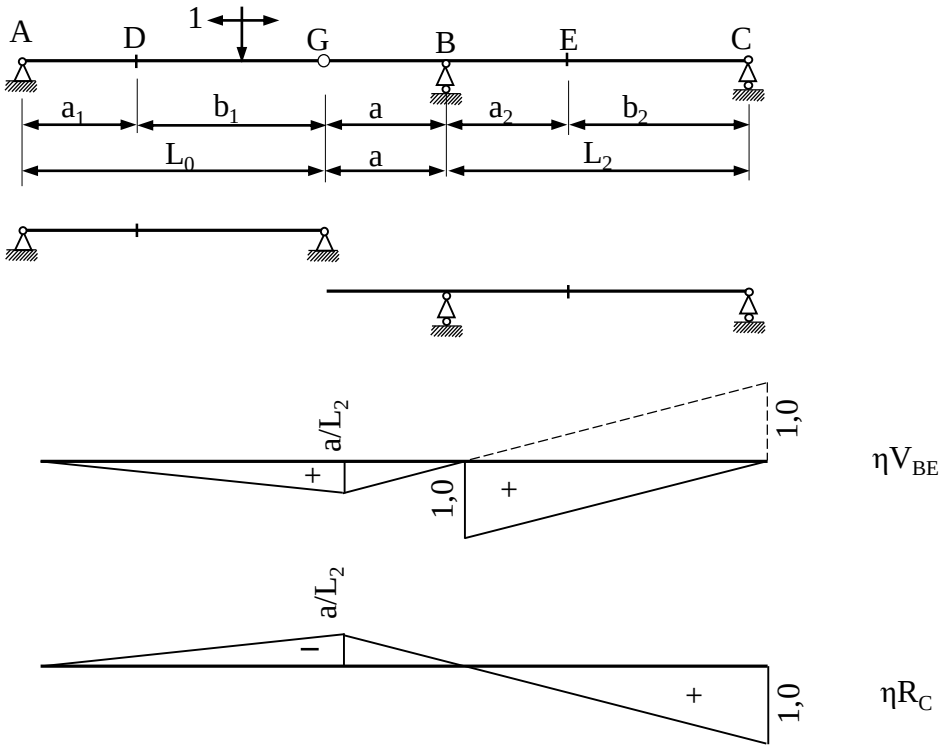
Influence line of shear force at E (V_E):

- The ordinate is zero only if the unit load is at A, B or C.
- First, the influence line of V_E for the simply supported beam of length L_2 is drawn, and then the ordinate $-a_2/L_2$ at E is connected to the zero ordinate at B and the line is extended until G.
- The ordinate at G is a/L_2 and it is connected to the zero ordinate at A.

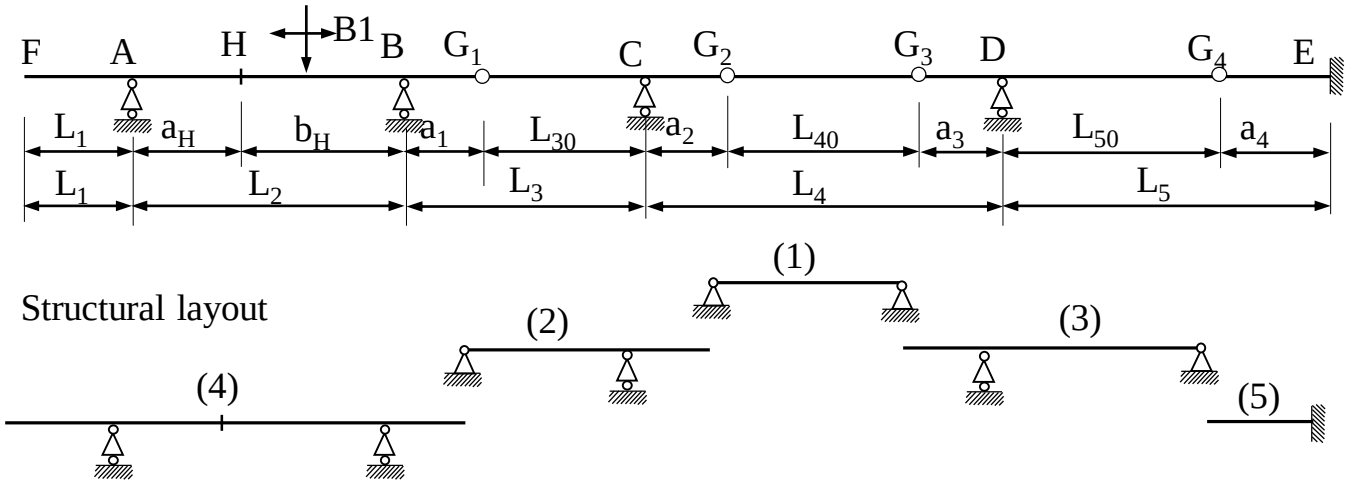


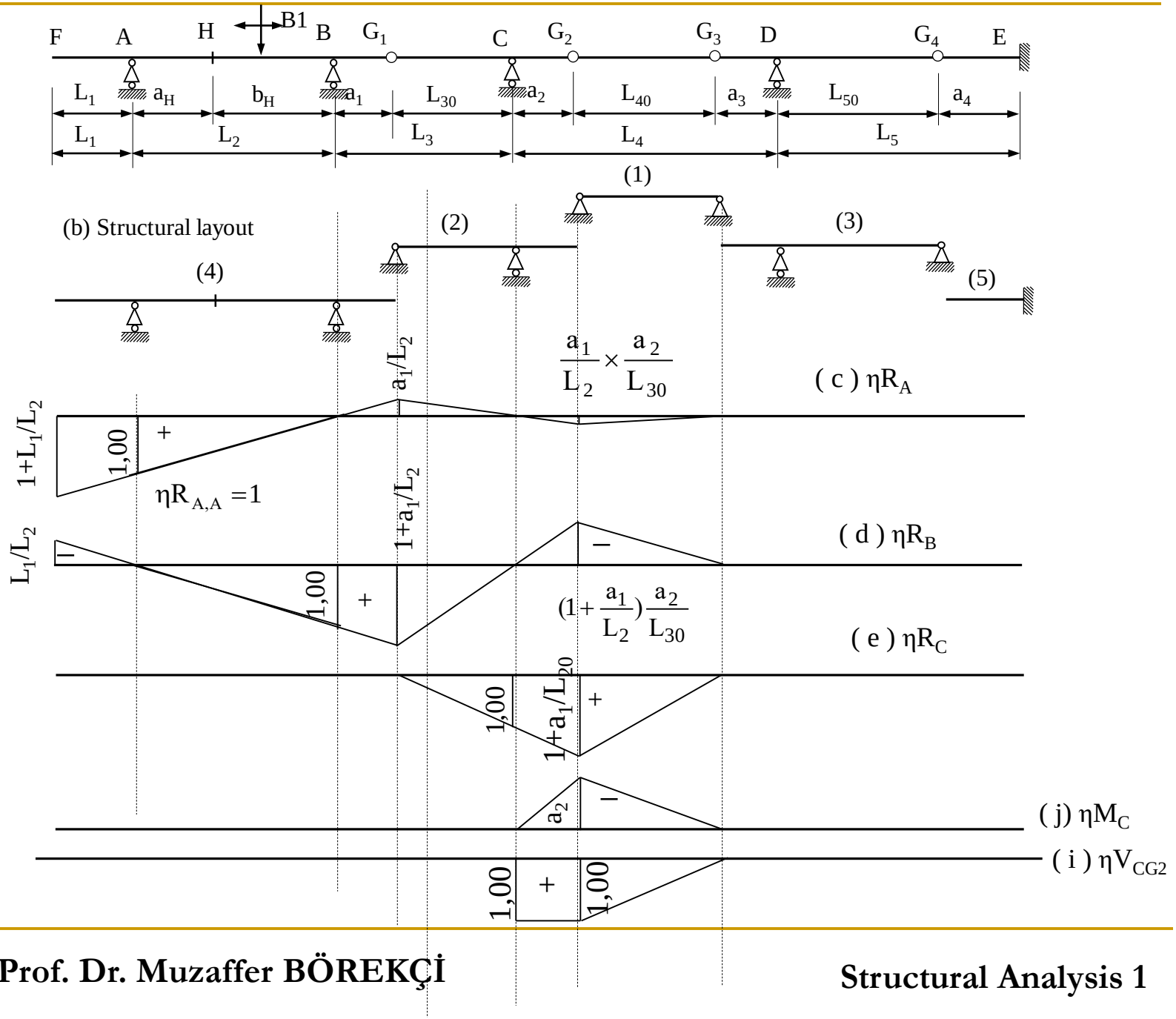
Influence line of bending moment at E (M_E):

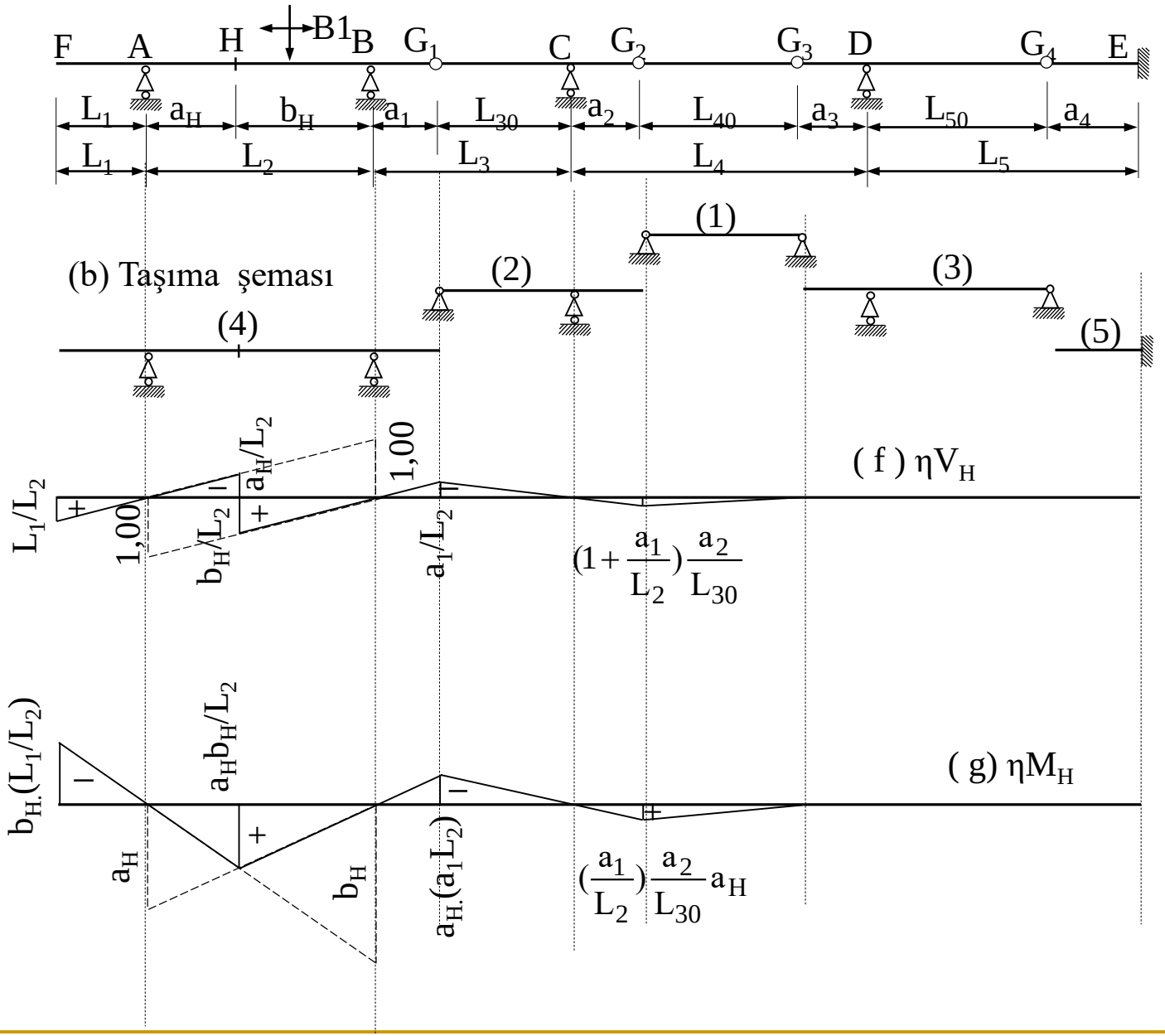
- The ordinate is zero only if the unit load is at A, B or C.
- First, the influence line of M_E for the simply supported beam of length L_2 is drawn. The ordinate at E is a_2b_2/L_2.
- The line between E and B is extended to G, where the ordinate is b_2a/L_2.
- The ordinate at G is connected to the zero ordinate at A.

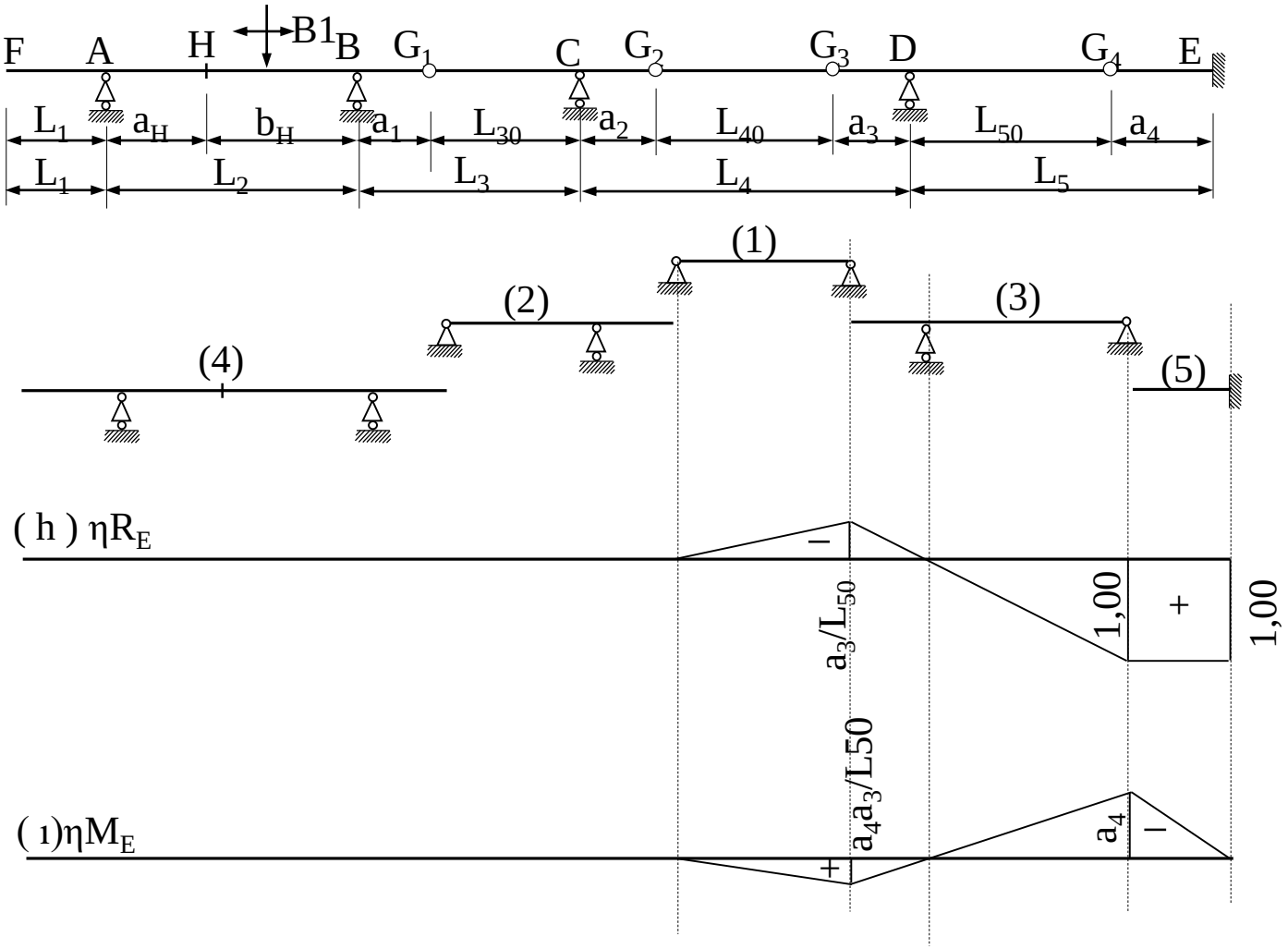

A
D
1
G
B
E
C
a1
b1
a
a2
b2
L0
a
L2
a/L2
1,0
+
1,0
+
ηVBE
a/L2
−
+
1,0
ηRC

Example 2:

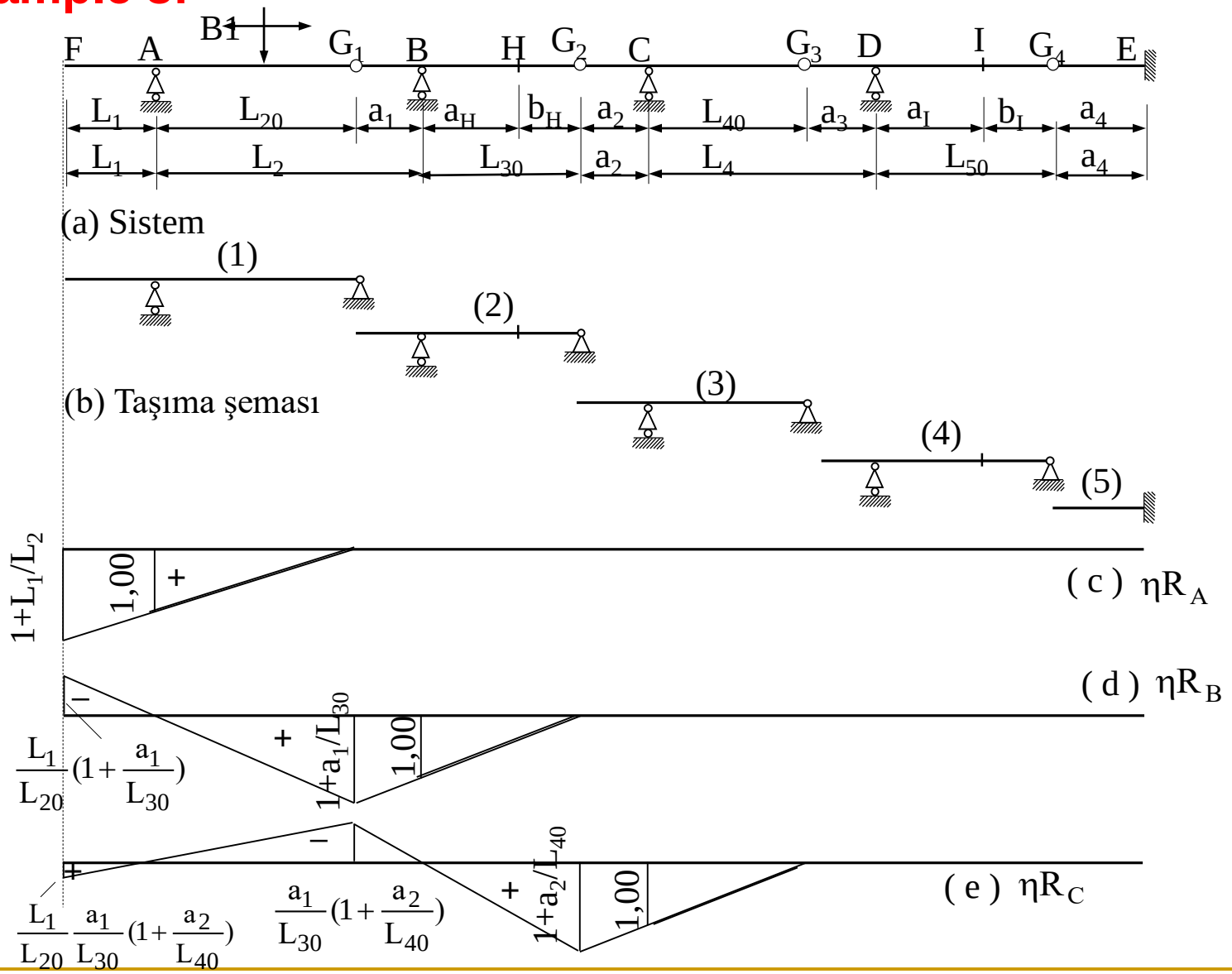

F
A
H
B1
B
G1
C
G2
G3
D
G4
E
L1
aH
bH
a1
L30
a2
L40
a3
L50
a4
L1
L2
L3
L4
L5
Structural layout
(1)
(2)
(3)
(4)
(5)

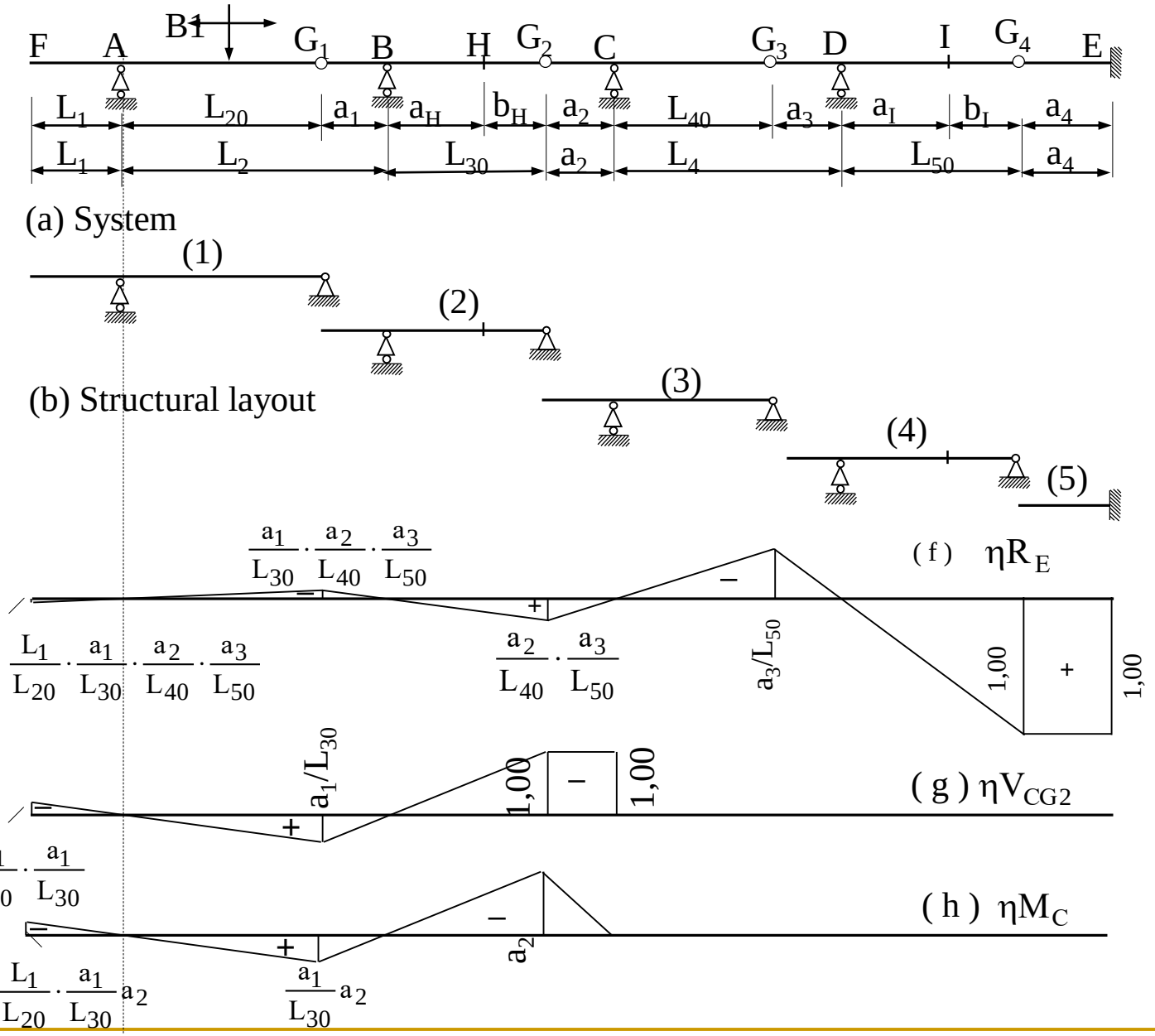

F
A
H
B1
B
G1
C
G2
G3
D
G4
E
L1
aH
bH
a1
L30
a2
L40
a3
L50
a4
L1
L2
L3
L4
L5
(b) Structural layout
(1)
(2)
(3)
(4)
(5)
a1/L2
a1/L2 × a2/L30
(c) ηRA
1+L1/L2
1,00
+
ηRA,A =1
1+a1/L2
L1/L2
−
(d) ηRB
1,00
+
(1+a1/L2) a2/L30
(e) ηRC
1,00
1+a1/L20
+
a2
−
(j) ηMC
(i) ηVCG2
1,00
+
1,00

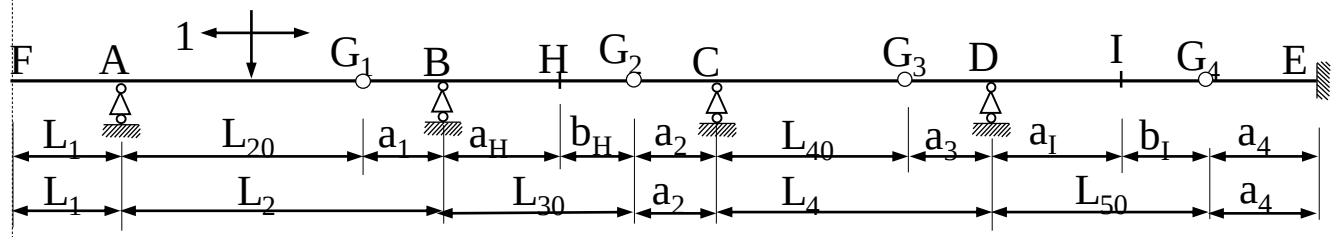

F
A
H
B1
B
G1
C
G2
G3
D
G4
E
L1
aH
bH
a1
L30
a2
L40
a3
L50
a4
L1
L2
L3
L4
L5
(b) Taşıma şeması
(1)
(2)
(3)
(4)
(5)
(f) ηVH
L1/L2
1,00
aH/L2
bH/L2
1,00
a1/L2
(1+a1/L2)a2/L30
(g) ηMH
bH.(L1/L2)
aHbH/L2
aH
bH
aH.(a1L2)
(a1/L2)a2/L30 aH


F
A
H
B1
B
G1
C
G2
G3
D
G4
E
L1
aH
bH
a1
L30
a2
L40
a3
L50
a4
L1
L2
L3
L4
L5
(1)
(2)
(3)
(4)
(5)
(h) ηRE
a3/L50
1,00
1,00
(ı)ηME
a4a3/L50
a4

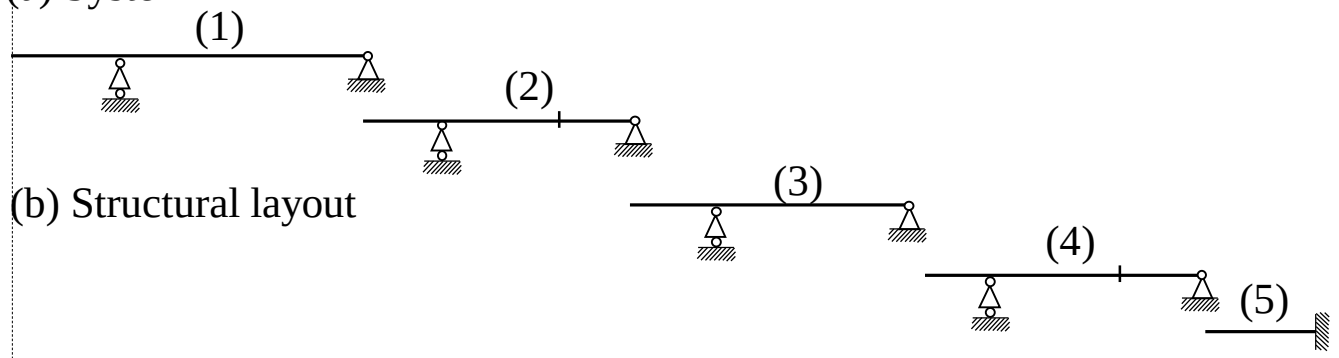
Example 3:


F
A
B1
G1
B
H
G2
C
G3
D
I
G4
E
L1
L20
a1
aH
bH
a2
L40
a3
aI
bI
a4
L1
L2
L30
a2
L4
L50
a4
(a) Sistem
(1)
(2)
(3)
(4)
(5)
(b) Taşıma şeması
1+L1/L2
1,00
+
(c) ηRA
(d) ηRB
L1/L20 (1+a1/L30)
+
1+a1/L30
1,00
(e) ηRC
L1/L20 a1/L30 (1+a2/L40)
a1/L30 (1+a2/L40)
+
1+a2/L40
1,00

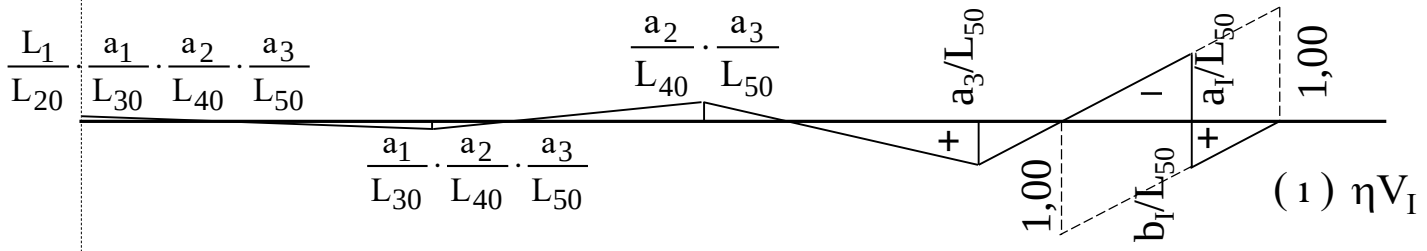

F
A
B1
G1
B
H
G2
C
G3
D
I
G4
E
L1
L20
a1
aH
bH
a2
L40
a3
aI
bI
a4
L1
L2
L30
a2
L4
L50
a4
(a) System
(1)
(2)
(3)
(4)
(5)
(b) Structural layout
(f) ηRE
(g) ηVCG2
(h) ηMC
1,00
1,00

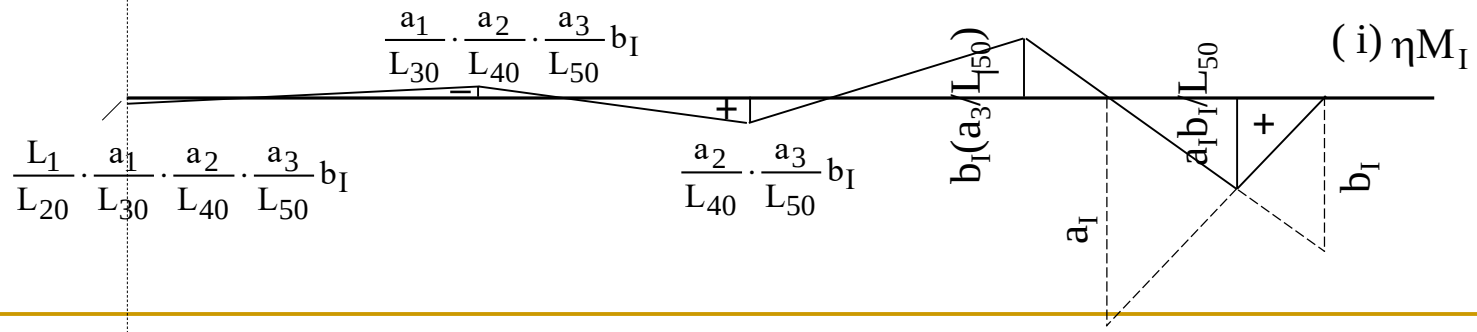


(a) System



(b) Structural layout





Reference:

All the figures have been taken from the book given below:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ